Mathematics for Risk Management in the Forex Market

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Abstract

This paper develops a mathematical framework for risk management in the Forex market, with emphasis on leverage, margin exposure, stop-loss and take-profit constraints, and the aggregation of multiple open positions into a single global profit or loss target.

Closed-form expressions are derived for position sizing, profit/loss margin estimation, and the final price levels required to achieve predefined outcomes across portfolios of buy and sell orders.

The proposed methodology is intended for systematic trading applications and provides practical tools for computational risk control in leveraged currency markets.

Keywords: Forex, Risk Management, Leverage, Margin, Stop-Loss, Take-Profit, Portfolio Risk, Quantitative Trading

JEL Classification: G11, G32, C02, C58

Disclaimer

The calculations and methods presented in this paper are for academic and educational purposes only. They do not constitute financial advice, investment recommendations, or guarantees of profitability. Trading in leveraged markets involves substantial risk, including the possibility of total capital loss.

Introduction

Risk management is a fundamental component of trading in leveraged financial markets, particularly in foreign exchange (Forex), where small price movements may generate significant gains or losses.

This paper proposes a mathematical approach to margin-based risk control, providing closed-form formulas for leverage exposure, stop-loss and take-profit levels, lot sizing, and global profit/loss targets across multiple open positions.

The objective is to provide practical tools that can be implemented algorithmically in systematic trading systems.

1 Leverage

Leverage is the loan your broker gives you to trade with more money than you actually have. It allows you to open a large position with a small amount of capital. For example, with 1:10 leverage on $100, you can trade as if you had 10 times that amount, or $1,000. But just as you can increase the loan to trade with more capital, you also increase the risk of loss. The higher the leverage, the closer you are to the liquidation point or total loss. That's why when trading with a certain amount of capital, you don't trade with the entire amount, but rather with a percentage. In practice, it's advisable not to use more than 2% of your total capital per position when trading in the Forex market. This will be explained more clearly in the following sections. For example, if we want to open a buy position on the EURUSD pair at a price of 1.12345 with leverage of 1:100, then the absolute loss of whatever amount is risked will be divided into 100 parts from the zero price to the buy price of 1.12345 and subtracted from that price, that is:

$$1.12345 - \frac{0 + 1.12345}{100} = 1.11221 \tag{1}$$

And the absolute profit is obtained by dividing the price by the leverage and adding it to that price, or like this:

$$1.12345 + \frac{0 + 1.12345}{100} = 1.13468 \tag{2}$$

From these calculations, we know that the absolute loss on the purchase occurs when the price falls from 1.12345 to 1.11221. And the absolute gain occurs when the price rises from 1.12345 to 1.13468. To generalize this into a formula and determine when there will be an absolute gain or loss, we use C_0 as the initial purchase price, which, according to the calculations in (1) and (2), is

$$P_c = C_0 \left(1 + \frac{r}{L}\right) \tag{3}$$

where P_c is the final purchase price for a profit or loss of r percent depending on the initial price C_0 and the leverage L.
For a sell position, the calculation is similar, using the following formula:

$$P_v = C_0 \left(1 - \frac{r}{L}\right) \tag{4}$$

2 Take Profit and Stop Loss

In the previous section, we saw how, for $r = 1$, we had an absolute loss or gain regardless of the amount risked. The expected profit or take profit (TP) is the profit expected from a buy or sell position, based on a proportion measured as a percentage of r or in pips. And the tolerated loss or stop loss (SL) is the loss that one limits to prevent further losses, also measured as a proportion of r or in pips. Equation (3) for a buy position is used to calculate the TP with $r > 0$ or to calculate the SL with $r < 0$. Similarly, equation (4) for a sell position is used to calculate the TP with $r < 0$ or to calculate the SL with $r > 0$. Let's look at some examples.

Example for a buy: Suppose we are convinced that the EURUSD price will rise from 1.10256 and therefore want to open a buy position with a 40% profit, a 20% allowable loss, and leverage of 1:400. Calculate the take profit (TP) and stop loss (SL) for this buy.

Solution: From the problem we have $C_0 = 1.10256$, $r_{tp} = 0.4$, $r_{sl} = -0.2$ and $L = 400$. Then to calculate TP according to equation (3) we have

$$P_{c,tp} = C_0 \left(1 + \frac{r_{tp}}{L}\right) = 1.10256 \left(1 + \frac{0.4}{400}\right) = 1.10366 \tag{5}$$

To calculate SL according to equation (3) we have

$$P_{c,sl} = C_0 \left(1 + \frac{r_{sl}}{L}\right) = 1.10256 \left(1 - \frac{0.2}{400}\right) = 1.10200 \tag{6}$$

So if we want to expect a 40% profit, we will have to wait for the price to rise to 1.10366, and if the price falls, we will have to close the position to avoid further losses, with a 20% tolerance when it reaches the value of 1.10200.

Example of a sell position: Suppose we are convinced that the EURUSD price will fall from 1.12526 and therefore want to open a sell position with an expected profit of 50%, a tolerated loss of 30%, and leverage of 1:500. Calculate the take profit (TP) and stop loss (SL) for this sell position.
Solution: From the problem, we have $C_0 = 1.12526$, $r_{tp} = 0.5$, $r_{sl} = -0.3$, and $L = 500$. Therefore, to calculate the TP according to equation (4), we have

$$P_{v,tp} = C_0 \left(1 - \frac{r_{tp}}{L}\right) = 1.12526 \left(1 - \frac{0.5}{500}\right) = 1.12413 \tag{7}$$

And to calculate the SL according to equation (4) we have

$$P_{v,sl} = C_0 \left(1 - \frac{r_{sl}}{L}\right) = 1.12526 \left(1 + \frac{0.3}{500}\right) = 1.12593 \tag{8}$$

Therefore, if we want to expect a 50% profit, we will have to wait for the price to drop to 1.12413, and if the price rises, we will have to close the position to avoid further losses, with a tolerance of 30% when it reaches the value of 1.12593.

3 Lot and Standard Lot

The lot size s is the trading volume for risking a buy or sell position, and the standard lot size γ is the base price of a currency pair when $s = 1$. For example, the standard lot size for EURUSD is $\gamma = 100,000$, meaning that for a lot size of $s = 1$, we have a contract to buy or sell 100,000 euros at the current dollar exchange rate against the euro. From this, we can have a trading volume of a mini lot worth 0.1, and a micro lot of 0.01. Generally, the lot size for forex varies between a minimum of 0.01 and a maximum of 200. For example, you can have a trading volume with a lot size of $s = 0.14$. But in dollar terms for the EURUSD pair, what is that lot size worth? That depends on the leverage and the initial trading price when a position is opened. The total amount m to risk is directly proportional to the lot size s, the standard lot size γ, and the initial price C_0, and inversely proportional to the leverage L. This is shown in the following formula:

$$m = \frac{s\gamma C_0}{L} \tag{9}$$

Example 1: We want to know how much to invest in a buy position of EURUSD with a lot size of 0.2 at a price of 1.13527 and leverage of 1:500.
Solution: According to the example problem, we have $s = 0.2$, $C_0 = 1.13527$, $\gamma = 100000$, and $L = 500$. Therefore, the amount to risk will be

$$m = \frac{s\gamma C_0}{L} = \frac{0.2 \cdot 100000 \cdot 1.13527}{500} = 45.41 \tag{10}$$

So, the investment amount will be \$45.41 for a lot size of 0.2 and leverage of 1:500 at a price of 1.13527.
In section 1, we mentioned that it's common to risk no more than 2% of total capital per trade. This is done to achieve scalable profits and keep losses under control in order to be consistent. So, how do I know which lot size to choose so that the amount equals that 2% of my total capital? Let's assume the total amount M is \$100 in my account, then 2% of \$100 is $0.02 \cdot 100 = \$2$. This is calculated with the following formula:

$$m = RM \tag{11}$$

where m is the amount to risk, R is the percentage to risk, and M is the total amount in our account. If we know the value to risk per trade m, but we don't know how much lot to choose, then from equation (9) we solve for s, and we will have

$$s = \frac{Lm}{\gamma C_0} \tag{12}$$

Example 2: If I have \$100 in my account and want to risk 2% on the EURUSD pair when its price is 1.11325 with a leverage of 1:500, then what will be the lot size or volume to open in that trade?
Solution: According to the problem, we have $L = 500$, $m = 100 \cdot 0.02 = 2$, and $C_0 = 1.1325$. Therefore, according to equation (12), we have

$$s = \frac{Lm}{\gamma C_0} = \frac{500 \cdot 2}{100000 \cdot 1.11325} = 0.008 \tag{13}$$

The fraction of the result in (13) is equivalent to 8 nano lots. However, this fraction does not exist unless the broker allows it. Typically, the minimum allowed lot size is 1 micro lot, or 0.01

of volume. Therefore, we must round the result of (13) to the nearest allowed lot size, that is, to $s = 0.01$. If we calculate the amount to risk according to this approximation using equation (9), we will have

$$m = \frac{0.01 \cdot 100000 \cdot 1.11325}{500} = 2.22 \tag{14}$$

The amount of \$2.22 approximates 2% of the total amount of \$100, which is allowed to open a buy or sell position. If one wants to earn exactly 2%, then the TP must be modified using formula (3) in section 1. The amount to risk also has a more technical name and is called margin, hence we use the letter m for the amount or margin in equation (9).

4 Profit and Loss Margin

We already know how to calculate the TP and SL of a position and what lot size to choose to risk a certain margin of our total account balance. Now we want to know how much profit margin we will have based on the TP price and the margin m to risk, and how much loss margin we will have based on the SL value. For the purchase, let's start from the concept that when a position is opened at the initial price C_0 with a certain lot size, we will have a leveraged amount, that is, as

$$Lm_0 = s\gamma C_0 \tag{15}$$

Then, if the price rises and we make a profit, at the final purchase price P_c the leveraged amount will be

$$Lm_c = s\gamma P_c \tag{16}$$

Therefore, the difference between equation (15) and (16) will be our profit, that is, as

$$G_c = Lm_c - Lm_0 = s\gamma P_c - s\gamma C_0 \tag{17}$$

or in factored form as

$$G_c = s\gamma(P_c - C_0) \tag{18}$$

But from equation (9) we have that

$$s\gamma = \frac{Lm}{C_0} \tag{19}$$

Replacing equation (19) in equation (18) we have

$$G_c = \frac{Lm}{C_0}(P_c - C_0) \tag{20}$$

which is

$$G_c = Am\left(\frac{P_c}{C_0} - 1\right) \tag{21}$$

For an expected sales margin of profit or loss, the procedure is similar and we have

$$G_v = Lm\left(1 - \frac{P_v}{C_0}\right) \tag{22}$$

or its equivalent as

$$G_v = s\gamma(C_0 - P_v) \tag{23}$$

Then with equation (21) for a purchase the profit or loss margin is calculated according to the TP or SL of P_c using equation (3) of section 1. The same for G_v of equation (22) for a sale: the profit or loss margin according to the TP or SL of P_v is calculated using equation (4) of section 1.

Example 1: From Example 1 in Section 3, if we expect a 45% profit and a 15% allowable loss, what will the profit and loss margins be according to those TP and SL?
Solution: Since it's a buy, we use equation (3) to find the price of that TP. And according to the data from Example 1 in Section 3, we have

$$P_{c,tp} = 1.13527\left(1 + \frac{0.45}{500}\right) = 1.13629 \tag{24}$$

And the price of the SL according to the problem data will be

$$P_{c,sl} = 1.13527\left(1 - \frac{0.15}{500}\right) = 1.13492 \tag{25}$$

According to the problem and the results of example 1 in section 3, the risk margin is $m = 45.41$, therefore the profit margin according to that value and the TP of the result in (24) using equation (21) will be

$$G_{c,tp} = Lm\left(\frac{P_{c,tp}}{C_0} - 1\right) = 500 \cdot 45.41 \cdot \left(\frac{1.13629}{1.13527} - 1\right) = 20.39 \tag{26}$$

If we used equation (18) we would have

$$G_{c,tp} = s\gamma(P_{c,tp} - C_0) = 0.2 \cdot 100000 \cdot (1.13629 - 1.13527) = 20.4 \tag{27}$$

We see that the results (26) and (27) are equal after rounding. Therefore, the reader can use either equation (21) or (18) depending on the problem data. This leads to the conclusion that the expected profit margin is \$20.40.
To calculate the loss margin based on the problem data and using the SL result in (25), we have

$$G_{c,sl} = Lm\left(\frac{P_{c,sl}}{C_0} - 1\right) = 500 \cdot 45.41 \cdot \left(\frac{1.13492}{1.13527} - 1\right) = -6.99 \tag{28}$$

If we use equation (18) we would have

$$G_{c,tp} = s\gamma(P_{c,tp} - C_0) = 0.2 \cdot 100000 \cdot (1.13492 - 1.13527) = -7 \tag{29}$$

We see that the results (28) and (29) are equal after rounding. Therefore, either equation (21) or (18) can be used, depending on the reader's preference and the problem data. This leads to the conclusion that the allowable loss margin is \$7.

5 Final Price of Multiple Purchases Based on Expected Margin

The calculations presented in this section and the following two sections are for illustrative and academic purposes only. They do not constitute financial advice nor do they guarantee results under actual market conditions.
Let's assume we have multiple open buy positions, but we don't want each one to reach its Stop Loss or Take Profit individually; instead, we want them to reach a profit or loss overall. Thus, we don't have multiple targets, but rather a single profit or loss target based on multiple buy positions.

From equation (21), we know that it is the formula for calculating the profit or loss of a single purchase. For the final profit of multiple purchases, we have the sum of all the profits and losses of the open positions, that is, as follows:

$$G_c = G_1 + G_2 + ... + G_n = \sum_{i=1}^{n} G_i = \sum_{i=1}^{n} Lm_i \left(\frac{P_c}{C_i} - 1 \right) \tag{30}$$

If we distribute the summation of equation (30) we have

$$\sum_{i=1}^{n} Lm_i \left(\frac{P_c}{C_i} - 1 \right) = \sum_{i=1}^{n} Lm_i \frac{P_c}{C_i} - \sum_{i=1}^{n} Lm_i \tag{31}$$

which is

$$G_c = \sum_{i=1}^{n} Lm_i \frac{P_c}{C_i} - \sum_{i=1}^{n} Lm_i \tag{32}$$

Rearranging the terms of equation (32) we have

$$L \sum_{i=0}^{n} m_i + G_c = LP_c \sum_{i=1}^{n} \frac{m_i}{C_i} \tag{33}$$

Solving for the variable P_c in equation (33) we will have

$$P_c = \frac{L \sum_{i=1}^{n} m_i + G_c}{L \sum_{i=1}^{n} \frac{m_i}{C_i}} \tag{34}$$

But from equation (9) we have

$$m_i = \frac{s_i \gamma C_i}{L} \tag{35}$$

Replacing equation (35) in equation (34) we have

$$P_c = \frac{\gamma \sum_{i=1}^{n} s_i C_i + G_c}{\gamma \sum_{i=1}^{n} s_i} \tag{36}$$

Both equation (34) and equation (36) are equivalent; only the type of variables used changes depending on the available data.

Example 1: Suppose we have three buy orders placed at different entry prices for the EURUSD pair. The first order was placed with a volume of 0.01 at a price of 1.08215, the second order with a volume of 0.02 at a price of 1.08032, and the third order with a volume of 0.03 at a price of 1.07629. If we want a profit of $20, calculate the take profit (TP) price.

Solution: According to the problem data we have

s_i	C_i
0.01	1.08215
0.02	1.08032
0.03	1.07629

Table 1: Values of s_i and C_i

From the values in Table 1, we will use equation (36). Replacing its values and considering that $G_c = 20$, we have

$$P_c = \frac{100000 \cdot (0.01 \cdot 1.08215 + 0.02 \cdot 1.08032 + 0.03 \cdot 1.07629) + 20}{100000 \cdot (0.01 + 0.02 + 0.03)} = 1.08194 \tag{37}$$

Therefore, the final price to obtain a \$20 profit from the 3 purchases will be when the EURUSD reaches a value of \$1.08194.

Example 2: Convert the purchases from Example 1 above to margins if the leverage is 1:500 and calculate according to equation (34), compare the result with Example 1.
Solution: From equation (35) and according to the prices from Example 1 with a leverage of $A = 500$, we have

$$\begin{aligned} m_1 &= \frac{s_1 \gamma C_1}{L} = \frac{0.01 \cdot 100000 \cdot 1.08215}{500} = 2.16430 \\ m_2 &= \frac{s_2 \gamma C_2}{L} = \frac{0.02 \cdot 100000 \cdot 1.08032}{500} = 4,32128 \\ m_3 &= \frac{s_3 \gamma C_3}{L} = \frac{0.03 \cdot 100000 \cdot 1.07629}{500} = 6,45774 \end{aligned} \tag{38}$$

From the results in (38) we have the following table of values:

m_i	C_i
2.16430	1.08215
4,32128	1.08032
6.45774	1.07629

Table 2: Values of m_i and C_i

Acording to the values in Table 2 and applying it to equation (34) with L_500 and $G_c = 0$ we will have

$$P_c = \frac{500 \cdot (2.16430 + 4,32128 + 6.45774) + 20}{500 \cdot \left(\frac{2.16430}{1.08215} + \frac{4,32128}{1.08032} + \frac{6.45774}{1.07629} \right)} = 1.08194 \tag{39}$$

From the result in (39) and this example, and the result in (37) from Example 1 above, we see that the result is the same. The reader can perform the calculations independently to verify the results. There is also the calculation when searching for a stop-loss or allowable loss for G_c. We leave it to the reader to perform exercises for allowable losses in this regard, using exercises similar to Examples 1 and 2 seen in this section.

6 Final Price of Multiple Sales Based on Expected Margin

Similar to the previous section, let's assume we have multiple open sell positions, but we don't want them all to reach their individual stop-loss or take-profit levels. Instead, we want them to reach a single profit or loss across multiple sell positions.
From equation (22), we know it's the formula for calculating the profit or loss of a single sale. The final profit from multiple sales is the sum of all the profits and losses from the open positions, or as follows:

$$G_v = G_1 + G_2 + ... + G_n = \sum_{i=0}^{n} G_i = \sum_{i=1}^{n} Lm_i \left(1 - \frac{P_v}{C_i}\right) \tag{40}$$

If we distribute the summation of equation (40) we have

$$\sum_{i=1}^{n} Lm_i \left(1 - \frac{P_v}{C_i}\right) = \sum_{i=1}^{n} Lm_i - \sum_{i=1}^{n} Lm_i \frac{P_v}{C_i} \tag{41}$$

which is

$$G_v = \sum_{i=1}^{n} Lm_i - \sum_{i=1}^{n} Lm_i \frac{P_v}{C_i} \tag{42}$$

Rearranging the terms of equation (42) we have

$$L \sum_{i=1}^{n} m_i - G_v = LP_c \sum_{i=1}^{n} \frac{m_i}{C_i} \tag{43}$$

Solving for the variable P_v in equation (43) we will have

$$P_v = \frac{L \sum_{i=0}^{n} m_i - G_v}{L \sum_{i=0}^{n} \frac{m_i}{C_i}} \tag{44}$$

But from equation (9) we have

$$m_i = \frac{s_i \gamma C_i}{L} \tag{45}$$

Replacing equation (45) in equation (44) we have

$$P_v = \frac{\gamma \sum_{i=0}^{n} s_i C_i - G_v}{\gamma \sum_{i=0}^{n} s_i} \tag{46}$$

Equation (44) and equation (46) are equivalent; only the type of variables used changes depending on the available data. We encourage the reader to perform calculations similar to those seen in the examples in the previous section to see how these formulas work.

7 Final Price of Multiple Purchases and Sales Based on Expected Margin

Let's assume we now have multiple open buy and sell positions, but we don't want them all to reach their individual stop-loss or take-profit levels. Instead, we want them to reach an overall profit or loss. In the previous two sections, we saw that this is possible when there are only buys or sells. In this section, we'll see how to combine both aspects to achieve a single profit or loss target across multiple positions.
From equations (21) and (22), for an overall profit or loss, let's designate the variable G_g, where this variable represents the sum of the buy and sell profits and losses.

$$G_g = (G_{c1} + G_{c2} + ... + G_{cn}) + (G_{v1} + G_{v2} + ... + G_{vn}) = \sum_{i=1}^{n_c} G_{ci} + \sum_{i=1}^{n_v} G_{vi} = G_c + G_v \tag{47}$$

But for G_c we know that its value is represented in equation (30), and similarly the value of G_v is represented by equation (40). If we substitute these values into equation (47), identifying each variable as either a purchase or a sale, we have

$$G_g = \sum_{i=1}^{n_c} Lm_{ci} \left(\frac{P_g}{C_{ci}} - 1 \right) + \sum_{i=1}^{n_v} Lm_{vi} \left(1 - \frac{P_g}{C_{vi}} \right) \tag{48}$$

Our goal here is to clear P_g. If we apply the distribution of summations to equation (48) we have

$$G_g = \sum_{i=1}^{n_c} Lm_{ci} \frac{P_g}{C_{ci}} - \sum_{i=1}^{n_c} Lm_{ci} + \sum_{i=1}^{n_v} Lm_{vi} - \sum_{i=1}^{n_v} Lm_{vi} \frac{P_g}{C_{vi}} \tag{49}$$

Factoring equation (49) according to the leverage L and P_g we have

$$G_g = LP_g \left(\sum_{i=1}^{n_c} \frac{m_{ci}}{C_{ci}} - \sum_{i=1}^{nv} \frac{m_{vi}}{C_{vi}} \right) - L \left(\sum_{i=1}^{n_c} m_{ci} - \sum_{i=1}^{n_v} m_{vi} \right) \tag{50}$$

Rearranging the terms of equation (50) we have

$$L \left(\sum_{i=1}^{n_c} m_{ci} - \sum_{i=1}^{n_v} m_{vi} \right) + G_g = LP_g \left(\sum_{i=1}^{n_c} \frac{m_{ci}}{C_{ci}} - \sum_{i=1}^{nv} \frac{m_{vi}}{C_{vi}} \right) \tag{51}$$

Solving for the variable P_g in equation (51) we will have

$$P_g = \frac{L \left(\sum_{i=1}^{n_c} m_{ci} - \sum_{i=1}^{n_v} m_{vi} \right) + G_g}{L \left(\sum_{i=1}^{n_c} \frac{m_{ci}}{C_{ci}} - \sum_{i=1}^{nv} \frac{m_{vi}}{C_{vi}} \right)} \tag{52}$$

But from equation (9) we know that

$$m_{ci} = \frac{s_{ci} \gamma C_{ci}}{L} \quad \wedge \quad m_{vi} = \frac{s_{vi} \gamma C_{vi}}{L} \tag{53}$$

Replacing the equations from (53) in equation (52) we have

$$P_g = \frac{\gamma\left(\sum_{i=1}^{n_c} s_{ci}C_{ci} - \sum_{i=1}^{n_v} s_{vi}C_{vi}\right) + G_g}{\gamma\left(\sum_{i=1}^{n_c} s_{ci} - \sum_{i=1}^{n_v} s_{vi}\right)} \tag{54}$$

Both equation (52) and equation (54) are equivalent; only the type of variable used changes depending on the available data. We leave it to the reader to perform calculations similar to those seen in the examples in section 5 to see how these formulas work.
The methods shown from section 5 to 7 are useful when you have a position with a negative margin. Thus, when you open a second buy/sell or more similar positions, we expect these positions to serve as a kind of redemption for the first one, which is experiencing a losing streak. For example, suppose we have a buy order, but the price falls instead of rising. This buy order is now incurring a loss. To achieve this, we open a second position, hoping it will offset the loss of the first. If the second position rises to a certain point in the asset's price between the first and second purchases, the profit will be zeroed out. However, if a positive profit margin is desired, we wait for the price to rise slightly above the zero profit point, so the difference between the profit from the second purchase and the loss from the first purchase is positive. This process can be applied to multiple purchases, sales, or a combination of both. This is most useful when programmed in a programming language to avoid performing all these calculations manually. Of course, the reader must know when to open a buy or sell position, or both, to apply these methods carefully. This depends on the investor's strategy.

8 Conclusion

This paper presented a mathematical framework for Forex risk management under leverage, deriving explicit formulas for position sizing, margin exposure, and profit/loss targeting.

In particular, closed-form expressions were obtained for aggregating multiple buy and sell positions into a single global take-profit or stop-loss objective, enabling systematic risk control in portfolio-level trading strategies.

Future work may include empirical validation with real market data, integration with probabilistic risk measures, and extensions to dynamic volatility regimes.

9 Python Implementation

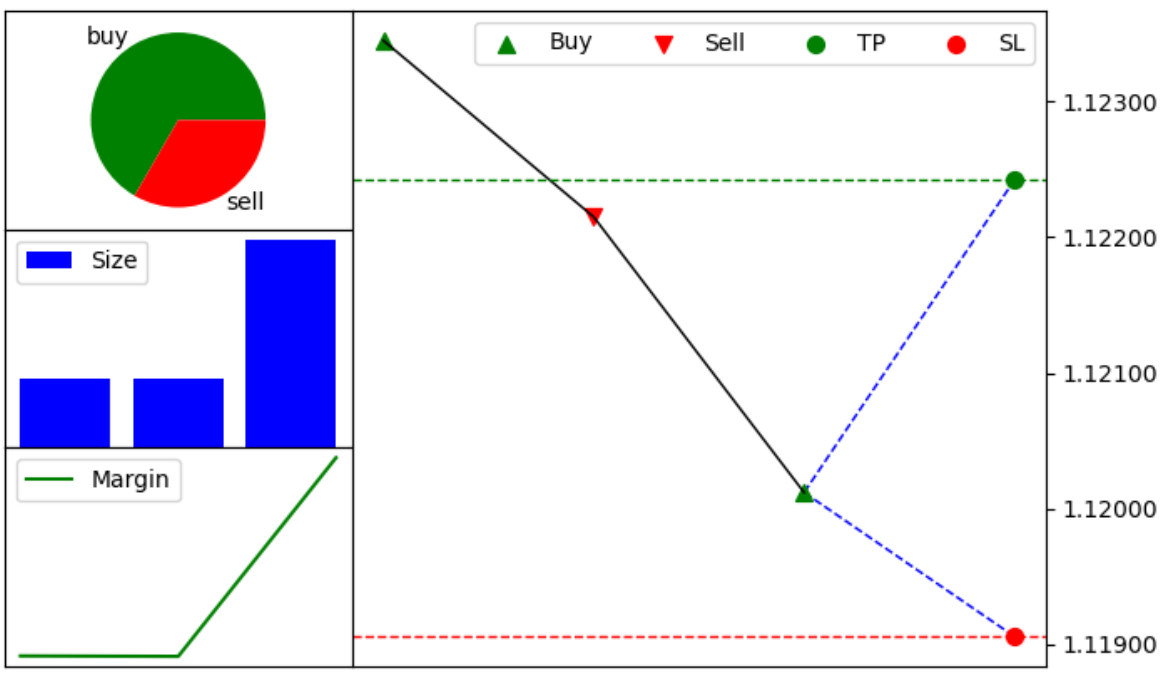
```python

import matplotlib.pyplot as plt
import matplotlib.ticker as mtick
import pandas as pd
import numpy as np

class risk_management:

    def __init__(self,leverage,standar_lot):
        self.leverage=leverage
        self.standar_lot=standar_lot
        self.positions=pd.DataFrame(columns=['Type','Volume','Margin','Price','TP','SL'])

    def margin_position(self,volume,price):
        return volume*self.standar_lot*price/self.leverage

    def position(self,type,volume,price,tp,sl):
        margin=self.margin_position(volume,price)
        if type=='buy':
            TP=price*(1+tp/self.leverage) if tp!=0 else 0.0
            SL=price*(1-sl/self.leverage) if sl!=0 else 0.0
        elif type=='sell':
            TP=price*(1-tp/self.leverage) if tp!=0 else 0.0
            SL=price*(1+sl/self.leverage) if sl!=0 else 0.0
        pos=[type,volume,margin,price,TP,SL]
        self.positions.loc[len(self.positions)]=pos

    def enter_positions(self):
        n=int(input('\nNumber of positions: '))
        for idx in range(n):
            print(f'\nEnter position {idx+1}:')
            type=input('\t Type: ')
            vol=float(input('\t Volume: '))
            price=float(input('\t Price: '))
            TP=float(input('\t TP percent: '))
            SL=float(input('\t SL percent: '))
            self.position(type,vol,price,TP,SL)
        return self.positions



    def show_positions(self,positions):
        self.positions['Margin']=self.positions['Margin'].round(2)
        positions[['TP','SL']]=self.positions[['TP','SL']].round(5)
        print('\n',positions)

    def count_positions(self,positions):
        total_positions=positions.loc[positions['Type'].isin(['buy','sell']),'Type']
        total_positions=total_positions.value_counts()
        count_buys=total_positions.get('buy',0)
        count_sells=total_positions.get('sell',0)
        return count_buys,count_sells

    def total_VolumeMargin(self,positions):
```

```
        sum=lambda pos,col: positions.loc[positions['Type']==pos,col].sum()
        volume_buy,volume_sell=sum('buy','Volume'),sum('sell','Volume')
        margin_buy,margin_sell=sum('buy','Margin'),sum('sell','Margin')
        return volume_buy,volume_sell,margin_buy,margin_sell

    def global_TpSl(self,vol_buy,vol_sell,m_buy,m_sell,positions,global_tp_per,global_sl_per):
        total_margin=m_buy+m_sell
        positions['mA']=positions['Volume']*positions['Price']*self.standar_lot
        sum_mA=lambda pos: positions.loc[positions['Type']==pos,'mA'].sum()
        num_global=lambda risk: sum_mA('buy')-sum_mA('sell')+total_margin*risk
        den_global=self.standar_lot*(vol_buy-vol_sell)
        global_tp=num_global(global_tp_per)/den_global if den_global!=0 else 0.0
        global_sl=num_global(-global_sl_per)/den_global if den_global!=0 else 0.0
        return global_tp,global_sl

    def multiple_positions(self,positions):
        count_buys,count_sells=self.count_positions(positions)
        print(f'\nTotal buys: {count_buys}')
        print(f'Total sells: {count_sells}')
        vol_buy,vol_sell,m_buy,m_sell=self.total_VolumeMargin(positions)
        print(f'Total volume: {round(vol_buy+vol_sell,2)}')
        print(f'Total margin: {round(m_buy+m_sell,2)}')
        global_tp_per=float(input('\nGlobal TP percent over total margin: '))
        global_sl_per=float(input('Global SL percent over total margin: '))
        g_tp,g_sl=self.global_TpSl(vol_buy,vol_sell,m_buy,m_sell,positions,
                                   global_tp_per,global_sl_per)
        print(f'Global TP: {round(g_tp,5)}')
        print(f'Global SL: {round(g_sl,5)}\n')

        data={'n_buys':count_buys,'n_sells':count_sells,'global_tp':g_tp,'global_sl':g_sl}

        return data

    def graph_positions(self,positions,data):
        fig=plt.figure(figsize=(7,4))
        fig.subplots_adjust(top=0.99,bottom=0.01,right=0.90,left=0.01,hspace=0,wspace=0)
        ax=fig.add_gridspec(3,3)

        ax0=fig.add_subplot(ax[0:1,0:1])
        ax1=fig.add_subplot(ax[0:1,0:1])    #pie n buys/sells
        ax2=fig.add_subplot(ax[1:2,0:1])    #bar volumes
        ax3=fig.add_subplot(ax[2:3,0:1])    #line margins
        ax4=fig.add_subplot(ax[:,1:3])      #line-scatter positions and tp/sl

        #PIE BUYS/SELLS
        n_buys,n_sells=data['n_buys'],data['n_sells']
        labels=['buy','sell']
        sizes=[n_buys,n_sells]
        colors=['green','red']
        ax0.set_xticks([])
        ax0.set_yticks([])
        ax1.clear()
        if len(positions)>0:
            ax1.pie(sizes,labels=labels,colors=colors)

        #BAR VOLUMES:
```

```
        ax2.clear()
        ax2.set_xticks([])
        ax2.set_yticks([])
        if len(positions)>0:
            ax2.bar(positions.index,positions['Volume'],color='blue',label='Size')
            ax2.legend(loc='upper left')

        #LINE MARINGS:
        ax3.clear()
        ax3.set_xticks([])
        ax3.set_yticks([])
        if len(positions)>0:
            if len(positions)==1:
                ax3.scatter(positions.index,positions['Margin'],color='g',label='Margin')
            else:
                ax3.plot(positions.index,positions['Margin'],c='green',label='Margin')
            ax3.legend(loc='upper left')

        #LINE SCATTER POSITIONS AND TP/SL:
        ax4.clear()
        ax4.yaxis.tick_right()
        ax4.set_xticks([])
        ax4.yaxis.set_major_formatter(mtick.FormatStrFormatter('%.5f'))
        if len(positions)>0:
            ax4.plot(positions.index,positions['Price'],c='black',lw=1)
            if data['n_buys']>0:
                buys=positions.loc[positions['Type']=='buy','Price']
                ax4.scatter(buys.index,buys,color='g',marker='^',s=50,label='Buy')
            if data['n_sells']>0:
                sells=positions.loc[positions['Type']=='sell','Price']
                ax4.scatter(sells.index,sells,color='r',marker='v',s=50,label='Sell')
            n_pos=len(positions)
            for g_risk,cc in zip(['global_tp','global_sl'],['g','r']):
                ax4.plot([n_pos-1,n_pos],[positions.at[n_pos-1,'Price'],data[g_risk]],
                    c='b',ls='--',lw=1,zorder=0)
                ax4.axhline(y=data[g_risk],color=cc,ls='--',lw=1)
            ax4.scatter([n_pos],[data['global_tp']],color='g',marker='o',s=50,label='TP')
            ax4.scatter([n_pos],[data['global_sl']],color='r',marker='o',s=50,label='SL')
            ax4.legend(loc='upper right',ncol=4)

        plt.show()


leverage,standar_lot=500,100_000
risk=risk_management(leverage,standar_lot)
positions=risk.enter_positions()
risk.show_positions(positions)
data=risk.multiple_positions(positions)
risk.graph_positions(positions,data)
```

The results of this code for 3 open positions are shown in the following images.

```
Number of positions: 3

Enter position 1:
         Type: buy
         Volume: 0.01
         Price: 1.12345
         TP percent: 0.1
         SL percent: 0.1

Enter position 2:
         Type: sell
         Volume: 0.01
         Price: 1.12215
         TP percent: 0.1
         SL percent: 0.1

Enter position 3:
         Type: buy
         Volume: 0.03
         Price: 1.12012
         TP percent: 0.1
         SL percent: 0.1

    Type  Volume  Margin    Price       TP       SL
0   buy    0.01    2.25  1.12345  1.12367  1.12323
1  sell    0.01    2.24  1.12215  1.12193  1.12237
2   buy    0.03    6.72  1.12012  1.12034  1.11990

Total buys: 2
Total sells: 1
Total volume: 0.05
Total margin: 11.21

Global TP percent over total margin: 0.5
Global SL percent over total margin: 0.4
Global TP: 1.12242
Global SL: 1.11906
```



References

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[2] John J. Murphy, *Technical Analysis of the Financial Markets*, New York Institute of Finance, 1999.

[3] Perry J. Kaufman, *Trading Systems and Methods*, Wiley, 6th edition, 2013.